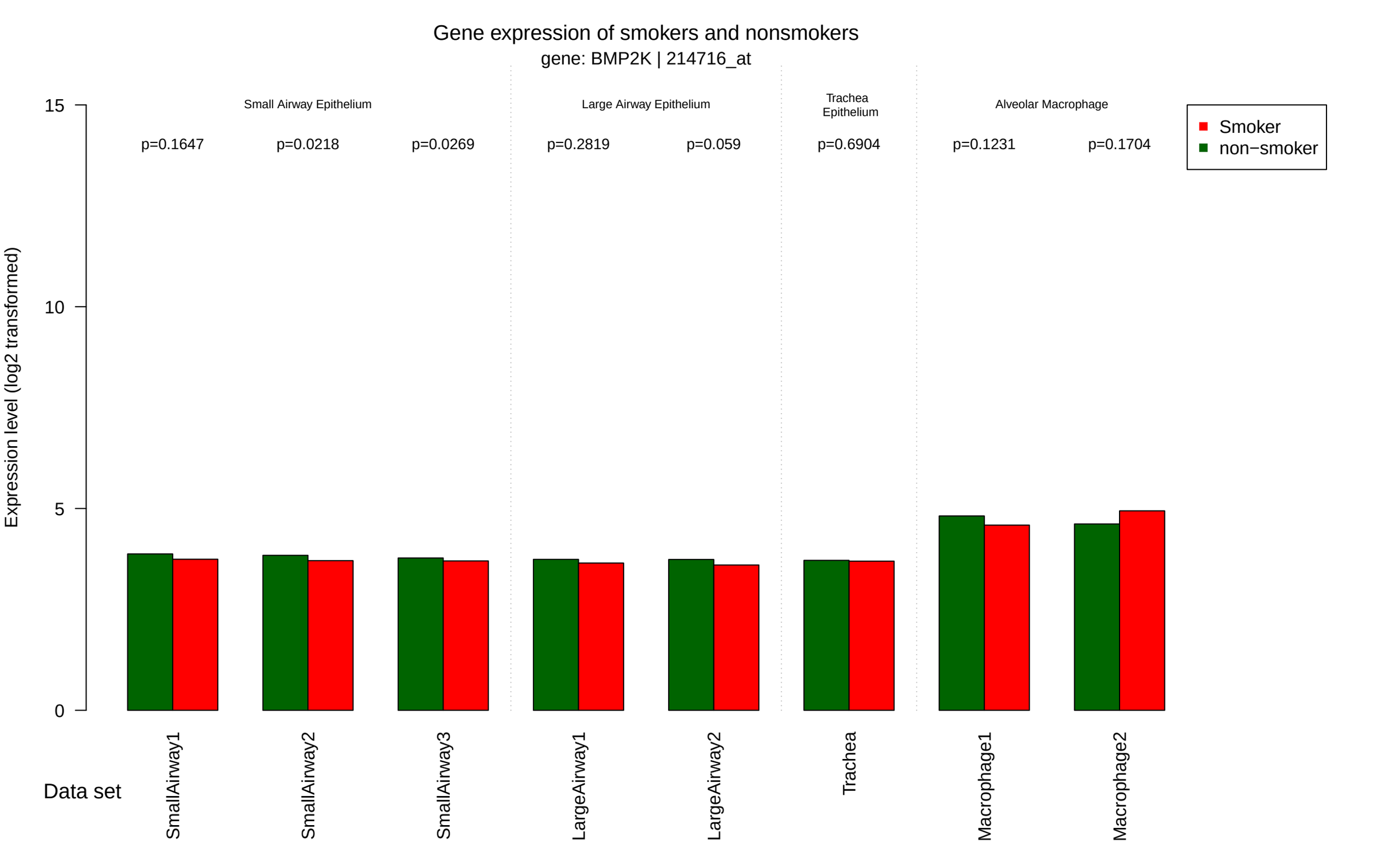
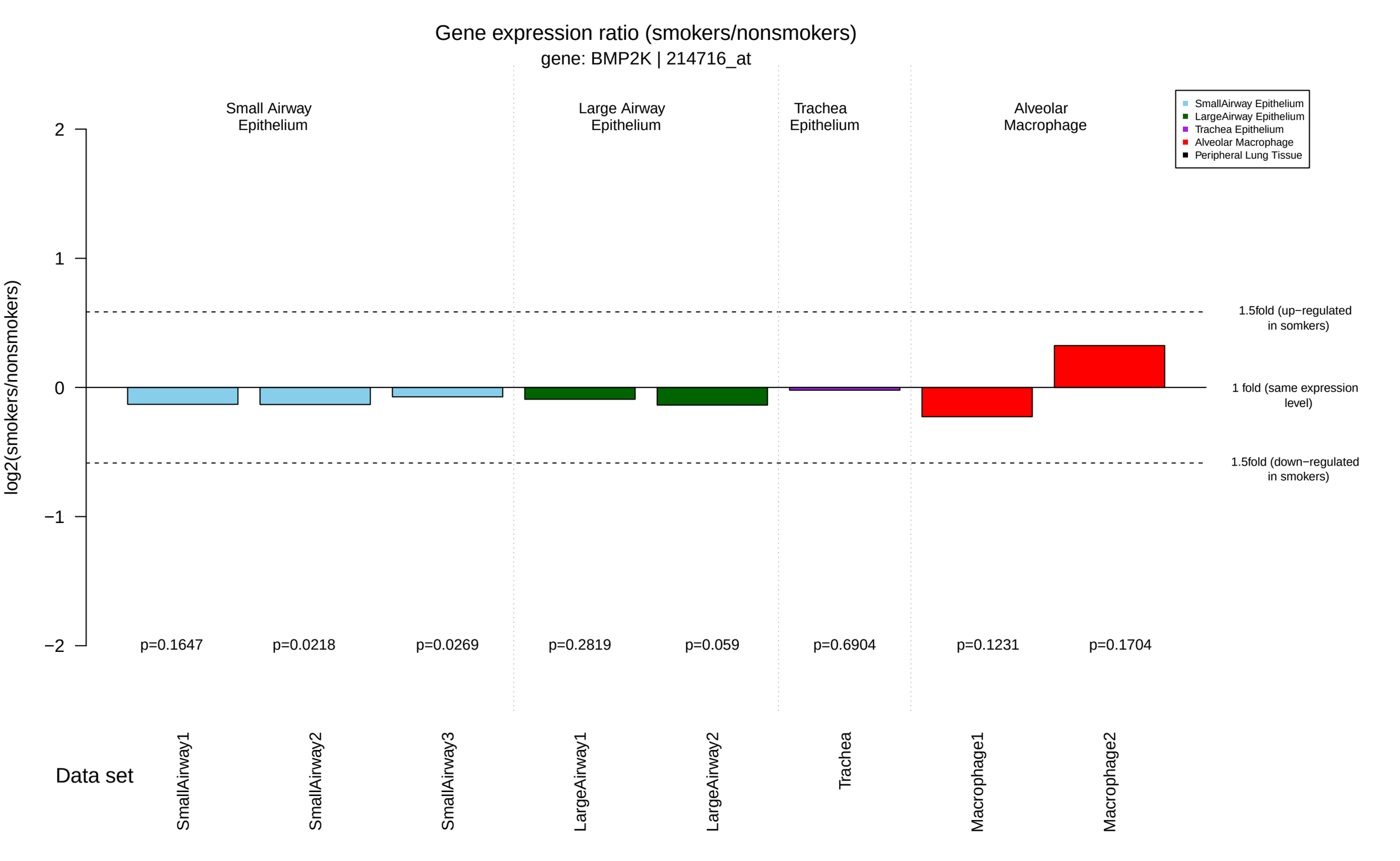
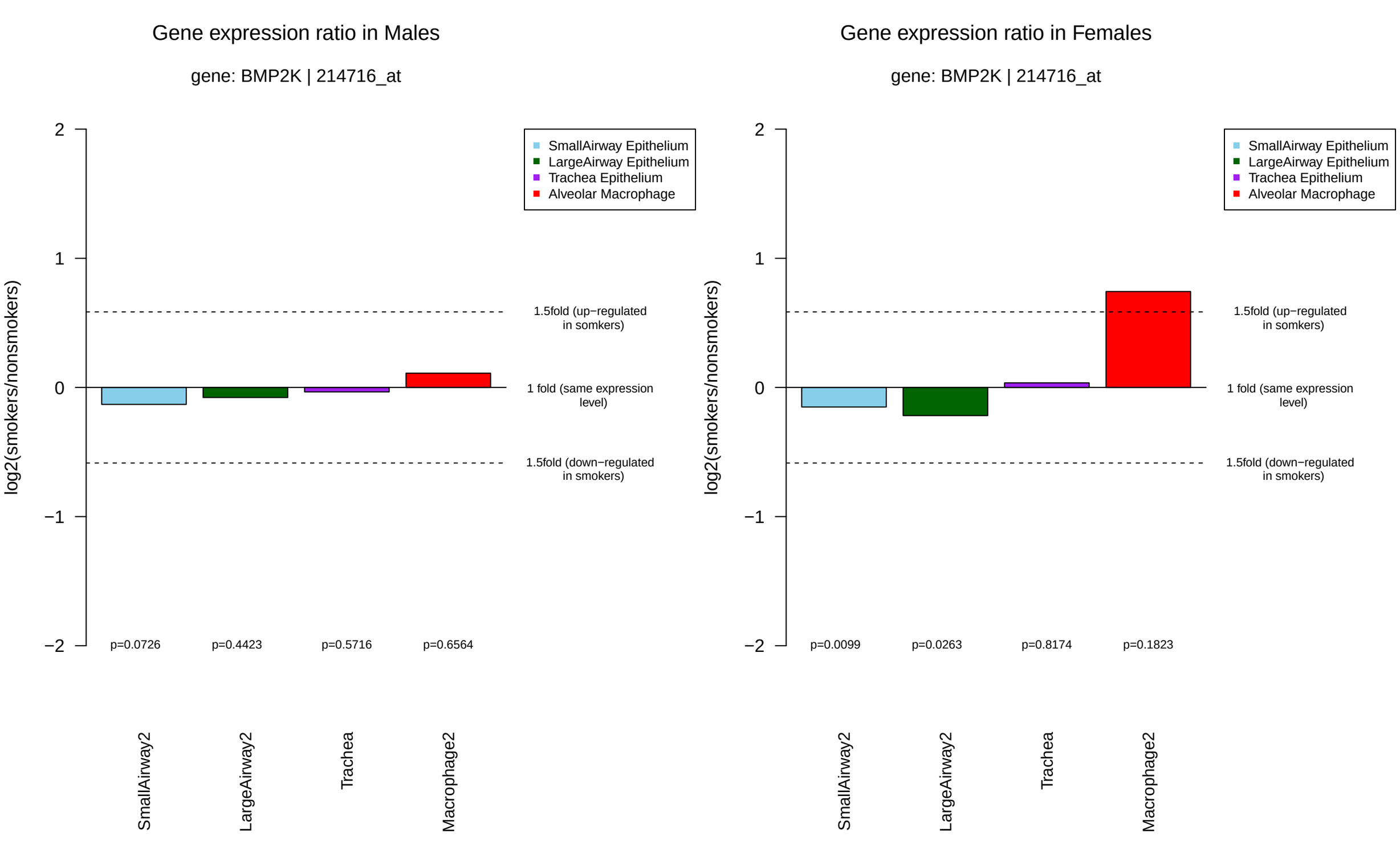
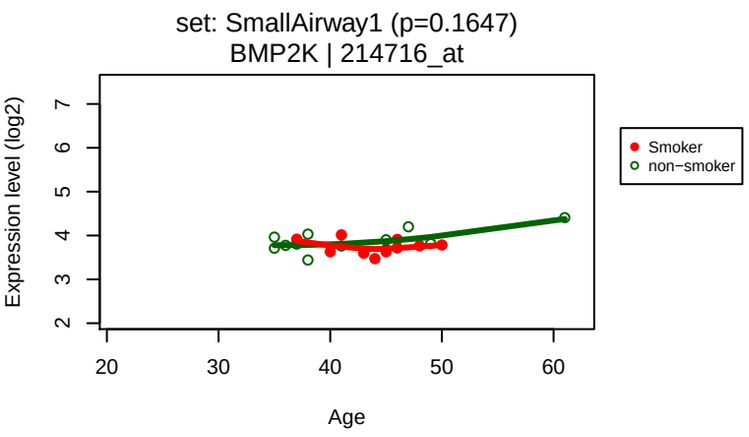
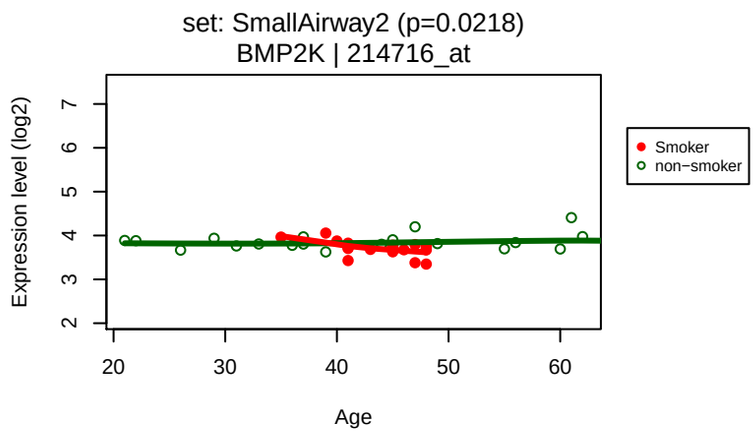
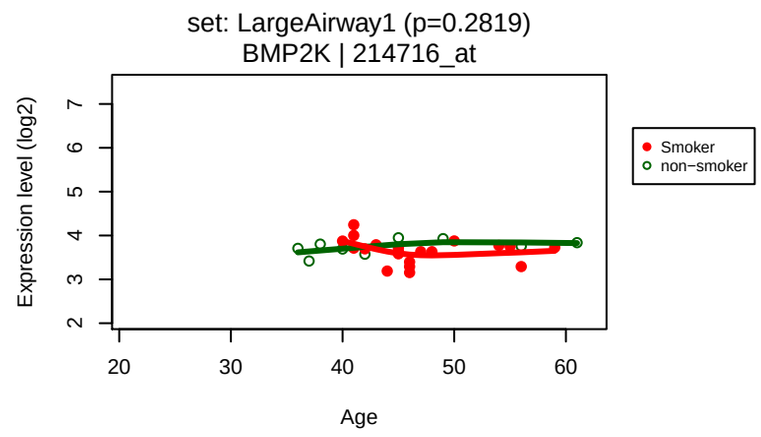

Gene expression of smokers and nonsmokers
gene: BMP2K | 214716_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1647
p=0.0218
p=0.0269
p=0.2819
p=0.059
p=0.6904
p=0.1231
p=0.1704
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

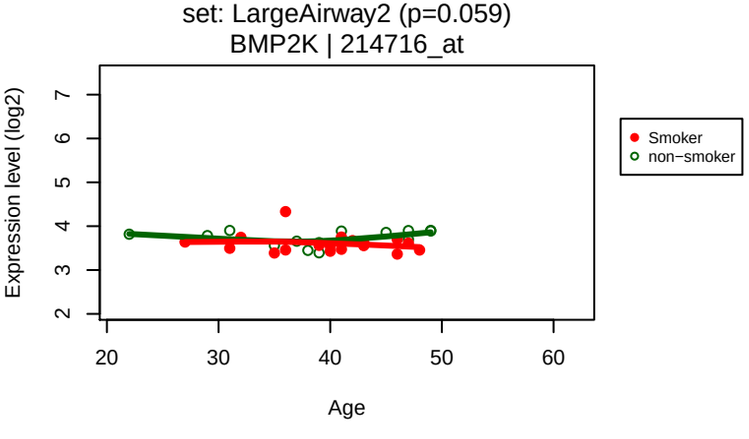

Gene expression ratio (smokers/nonsmokers)
gene: BMP2K | 214716_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1647
p=0.0218
p=0.0269
p=0.2819
p=0.059
p=0.6904
p=0.1231
p=0.1704
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

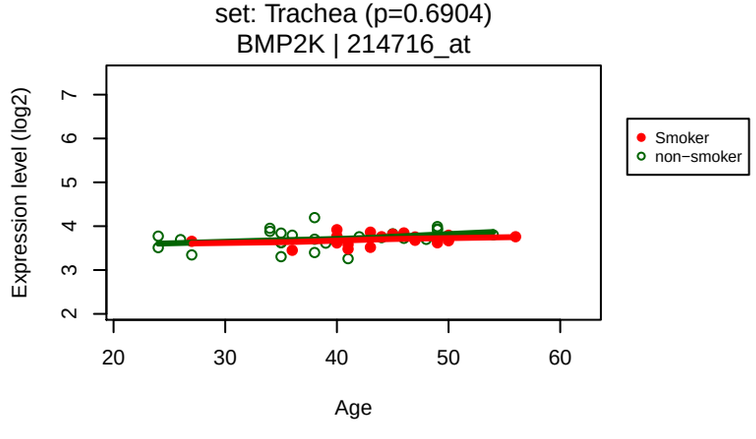

Gene expression ratio in Males
gene: BMP2K | 214716_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0726
p=0.4423
p=0.5716
p=0.6564
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2K | 214716_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0099
p=0.0263
p=0.8174
p=0.1823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

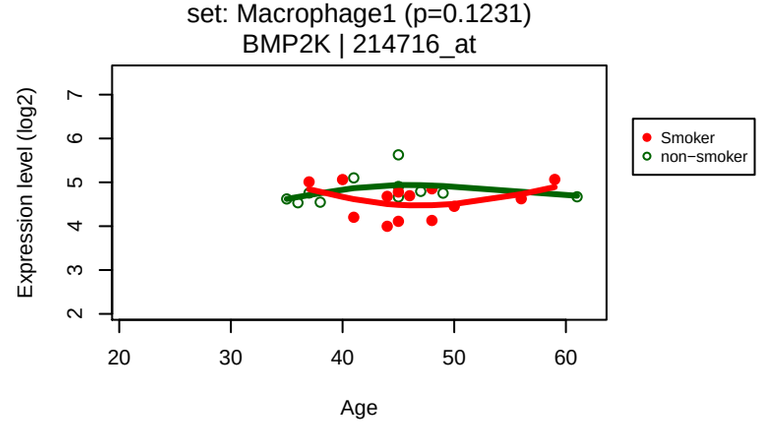

set: SmallAirway1 (p=0.1647)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker

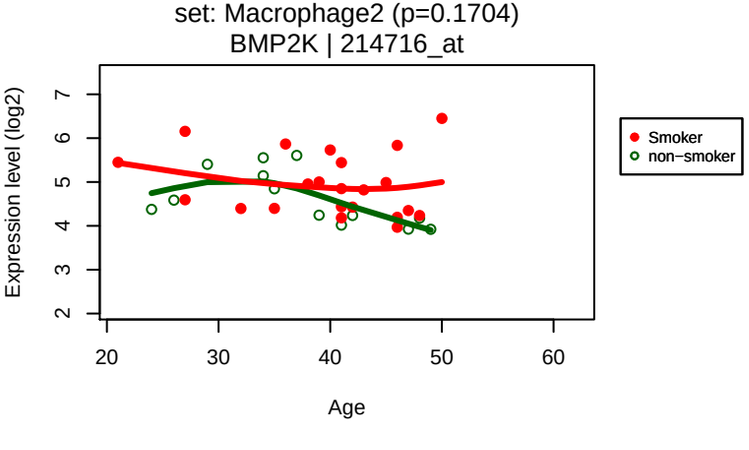

set: SmallAirway2 (p=0.0218)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker

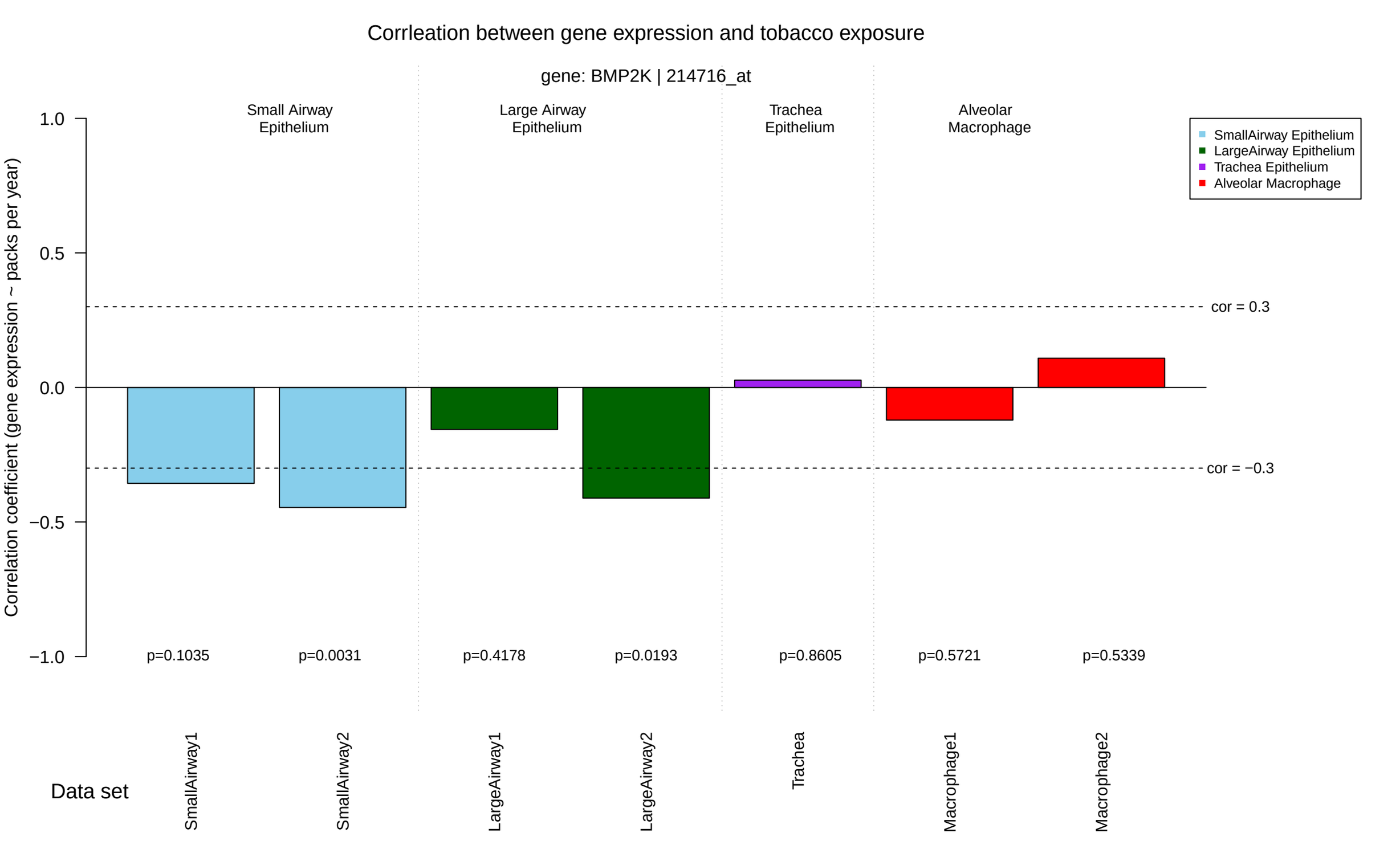

set: LargeAirway1 (p=0.2819)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.059)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6904)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1231)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1704)
BMP2K | 214716_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2K | 214716_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1035
p=0.0031
p=0.4178
p=0.0193
p=0.8605
p=0.5721
p=0.5339
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2